

WEEK 6 OUTPUT

HealthConnect Experience Lab - AnalystLab Africa

Week 5 Review

In Week 5, I prepared the HealthConnect data and conducted an initial exploration analysis to have an understanding of what I was working with. My key findings from the data were that the no show rate rested at 48.46% (2,423 of 5,000 appointments) and that 40% of patients (679 of 1,696) account for 72.4% of all no-shows. Also, for week 5, I didn't explore why there was a high percentage of no shows. I only studied what most of the factors seemed to suggest. For week 6, I had planned to deepen the reminder-effectiveness analysis and to build out the KPI dashboard for stakeholder review. Instead, I deepened the overall analysis to understand the appointment outcomes, and why the "no-shows"

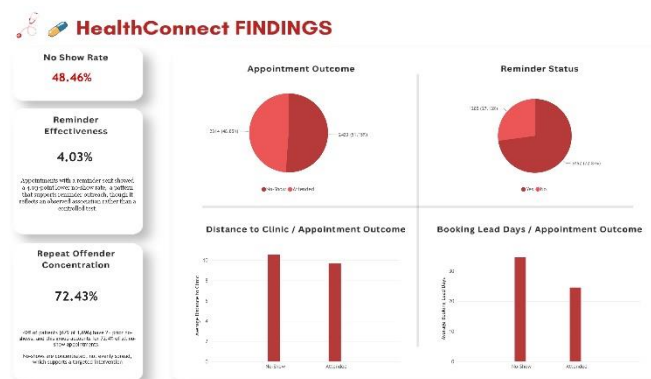
Week 6 Progress

To have an idea of what I worked on for this week, see the visualizations below.



This is what my dashboard looked like in Week 5. I only explored what the factors seemed to suggest.

Upon collaboration with a data science intern, this is what my dashboard was revised to:



Advanced Analysis Findings

1. Reminder effectiveness sits at just **4.03 percentage points**.

Appointments with a reminder still no-show at a high rate, meaning reminders are reaching the majority of patients but aren't moving the needle much on their own.

2. **72.43%** of no-shows come from patients with 2+ prior no-shows. It confirms no-show loss is concentrated, not distributed evenly across the patient base, and supports a targeted rather than blanket intervention.

3. The patients who attend their appointments live at least 9.67km from the facility. Those who are a no show live at least 10.53km away.
4. Booking lead time has a visible effect on no shows. Most patients who have an average of 34 booking lead days, were more likely to not show up unlike the patients with at least 24 booking lead days.

My Recommendations Based on the Findings

1. Since the patients who live at 10.53km away contribute to the no show situation, the facility could try a mobile clinic for follow ups. Before the mobile clinic is out, reminders should be sent out to patients. Basically prioritizing a targeted intervention for the no-show patients rather than a blanket clinic-wide change.
2. The medical practitioners should have a cut-off on booking lead days. Days averaging past a month are most likely to be forgotten by patients, or in some cases, may result in a patient looking for a second opinion within a shorter time.
3. Track the three core KPIs (No-Show Rate, Reminder Lift, Repeat-Offender Concentration) on an ongoing dashboard to monitor whether interventions are working.

Cross-Track Collaboration: Integration Point

- I worked with the data science track.
- What I wanted from the track was to understand what other factors could be included in my analysis to make it stronger such that if I were to hand-off my findings to a different data scientist, they would know how to go around everything to build their models.
- What I provided to the track was the understanding that there is a data quality issue in `waiting_time_minutes` (some patients that were a no show or cancelled still had their waiting time included)

Why The Connection is Relevant: Data Analysis closely informs modeling on Data Science. I wanted to conduct a stronger analysis that would communicate strong insights to stakeholders by picking out tips from a data science intern.

Improvement from the Collaboration

My dashboard was updated to visuals that actually communicate the problem and my analysis shared a better report. The cancelled appointments were excluded from analysis because we're strictly focusing on no-show cases.

Limitation

Distance to clinic has 1.8% of blanks, a percentage that doesn't necessarily skew the results towards one side but worth noting.

Analytical Testing Requirements for Week 7

1. A chi-square test of independence between `reminder_sent` and `appointment_outcome`, to confirm the observed 4.03pp gap reflects a real relationship rather than sample variation.
2. A t-test comparing average `distance_to_clinic_km` and average `booking_lead_days` between No-Show and Attended groups, to confirm the gaps shown in the dashboard's bar charts are statistically meaningful.
3. Comparing the observed 72.4% concentration against what would be expected if no-shows were evenly distributed across patients, to confirm the pattern is meaningfully disproportionate rather than a natural result of random spread.